Scattered Spring: How Temperature Synchronizes Ecosystems and its Consequences for Climate Change

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Abstract

A wide variety of biological processes are triggered by cumulative temperatures. Many plant events, such as leafout and flowering, occur once the temperature sum exceeds a threshold. In this way, cumulative temperatures are thought to ~~may~~ coordinate the timing of ecological events both within a species and between the species that make up an ~~potentially~~ ecosystem. But a growing number of ~~increasing~~ reports show slowing responses to warming, increasing variance in the timing of spring events, and other anomalies, raising questions about the resilience of ~~how reliable this~~ temperature-based coordination to ~~is with~~ anthropogenic climate change. Researchers have proposed a suite of varying hypotheses to explain these findings, all of which add complexity beyond a simple model of events triggered by cumulative temperatures. Yet is such complexity necessary? ~~Yet this research has not fully evaluated predictions from the simple model.~~

We address this question ~~this gap~~ by modeling temperature-triggered biological processes as stopped random walks. Cumulative temperature determines the position of the walk, and the occurrence of the biological event is the stopping time. This framework yields two theory-driven implications for how warming alters the timing and synchrony of seasonal events. First, the relationship between temperature and the expected event time is nonlinear. As temperatures increase, the average springtime event such as flowering occurs earlier, but the rate at which they advance can diminish or even accelerate. Second, warming affects not only the expected time but its variability. As temperatures increase and springtime events shift from the equinox toward the solstice, springtime events become less coordinated, producing a more variable or "scattered" spring. We verify both implications empirically using experimental and observational data.

Introduction

Many biological events are triggered not by the temperature on a single day, but by the temperature accumulated over many days. A canonical example is the law of the flowering plants, which states that a bud blooms in the spring once cumulative temperatures exceed a plant-specific threshold. This and similar mechanisms synchronize a wide variety of seasonal events in ecosystems—for instance, ensuring that pollinators emerge when flowers are in bloom. They also support an array of forecasting tools in which scientists track cumulative temperature or 'growing degree days' to predict harvest dates, manage pest emergence, and assess how shifting temperature patterns reshape ecological communities with climate change.

Shifts to earlier biological events—i.e., earlier phenology or 'advancement'—has been the clearest fingerprint of anthropogenic climate change across both natural and managed systems. But these shifts have proven difficult to predict or explain. ~~have recently become less predictable.~~ In recent decades, the rate of advancement has decelerated in that biological events advance less with each degree of warming (fu2015declining). At the same time, ~~With this,~~ the variance in the timing of phenological events has in some cases declined and in others increased, without a clear pattern (vitasse2018global,stemkovski2023disorder). In particular, ~~At the same time,~~ previously unknown or rare phenological events appear more common (chuine2025living) (JA comment: Occurrence of previously rare events are a sign of increased variance, no?). While research has provided a myriad of explanations for these events—often with incoherent ~~directly contrasting~~ predictions and logic (fu2015declining,wolkovich2021simple,gao2024thermal)—no work has examined the extent to which these observations are consistent with the simple cumulative sum model. ~~predictions from the existing cumulative sum model fully.~~

Indeed, scientists have used a basic cumulative sum model for over two centuries, but have rarely if ever integrated the relevant mathematical tools developed to characterize its behavior—specifically results from renewal theory designed for analyzing stopped random walks. ~~Despite the shared mathematical structure, few have connected growing degree-day models and the modern limit theory of stopped random walks.~~ Here we draw on results from renewal theory to predict how events triggered by a cumulative temperature sum will shift with warming. Our model predicts observed advances as well as shifts in variance. We show predictions depend critically on characteristics of the underlying temperature regime. In particular, the variance in the timing of biological events can decrease or increase with warming depending on whether temperatures during the period of accumulation are stationary (as in the winter and summer near the solstice) or increasing (as in the spring near the equinox). Taken together our results show how climate change can (i) move spring events earlier at an increasing or decelerating rate, and (ii) increase the variance of their timings, producing less synchronized events as they arrive earlier, a phenomenon we refer to as a 'scattered spring.'

Results & Discussion

Many spring events occur when cumulative temperatures first pass or 'hit' a plant-specific temperature threshold. Hitting-time models fall within the domain of stochastic process theory, with which we can approximate the distribution of event timings (hitting times) and thus predict the behavior of temperature-triggered biological events.

Consider one such event, the bloom date of a bud. We model effective temperatures (X_i)—the temperatures experienced by a plant bud at a given location—with a baseline temperature (α), a within-season warming rate (β, commonly seen in the spring months, see Fig 1), and mean-zero noise (ϵ_i, thus $X_i = \alpha + \beta\, i + \epsilon_i$,). Noise in this formulation could represent many biological aspects depending on the scale and our developing understanding of what triggers spring plant events. For example, in our focus on the bud-level it represents idiosyncratic variation at that bud, which could be driven potentially by microclimate variation (schwartz2014separating), ~~including~~ bud color variation (peaucelle2022accurate), or internal differences that vary when the bud starts accumulating temperatures (CITESmolecularchillpapers). Bloom occurs on the day (n) that the cumulative temperature ($Z_n = \sum_{i=1}^{n} X_i$) first passes or 'hits' a plant-specific tempearture-sum threshold t > 0, i.e. $\nu(t) = \min\{m : Z_m > t\}$). Thus $\nu(t)$ is the bloom date, the day at which cumulative temperatures first hits the threshold.

Standard results for stopped random walks yield asymptotic normal approximations for $\nu(t)$ when t is sufficiently large that accumulation takes place over many days. ~~The behavior of the asymptotic distribution is governed by the parameters α and β. A central implication of this framework is that predictions depend strongly~~ The mean and variance of the approximation depend crucially on the underlying temperature regime (i.e., values of α and β) during which the plant accumulates temperature. In the stationary regime, average temperatures are relatively flat during accumulation ($\beta \approx 0$, common in winter months near the solstice), and cumulative temperature grows at an approximately linear rate; thus the expected bloom date advances with climate change in step with increasing baseline temperatures ($\mathbb{E}[\nu(t)] \approx \frac{t}{\alpha}$) as is often expected in the literature (vitasse2022great). In the non-stationary regime, average temperatures increase during accumulation ($\beta > 0$, common in spring months closer to the equinox), and cumulative temperatures grow quadratically. In this regime, bloom date advances as a function of the rate of within-season warming ($\mathbb{E}[\nu(t)] \approx \sqrt{\frac{2t}{\beta}}$)—that is, how quickly spring temperatures increase.

In both regimes, the relationship between bloom dates and temperature is non-linear as observed ~~increasing temperatures advance bloom dates in a non-linear function, similar to the shifts described~~ in recent papers (fu2015declining,wolkovich2021simple). In addition, in both regimes, increasing temperatures advance bloom dates at a diminishing rate as warming (measured by the baseline temperature, α or within-season rate, β) increases relative to the size of the threshold (for the stationary regime: $\frac{\partial}{\partial \alpha}\,\mathbb{E}[\nu(t)] \approx -\frac{t}{\alpha^2}$; for the non-stationary regime: $\frac{\partial}{\partial \beta}\,\mathbb{E}[\nu(t)] \approx -\frac{1}{2}\sqrt{\frac{2t}{\beta^3}}$). But when climate change shifts the period when plants accumulate temperatures from the non-stationary regime that holds near the spring equinox to the more stationary regime that holds near the winter solstice, advances may actually increase, particularly if increases in the baseline (α) outpace increases in the within-season rate (β).

Importantly, these two temperature regimes make very different predictions regarding the synchrony of events—that is the variance in event timings within the same year and location. Both predict decreased variance with warming (for the stationary regime: $\mathrm{Var}(\nu(t)) \approx \frac{\sigma^2\, t}{\alpha^3}$; for the non-stationary regime: $\mathrm{Var}(\nu(t)) \approx \frac{\sigma^2}{\beta^{3/2}\sqrt{2t}}$), but

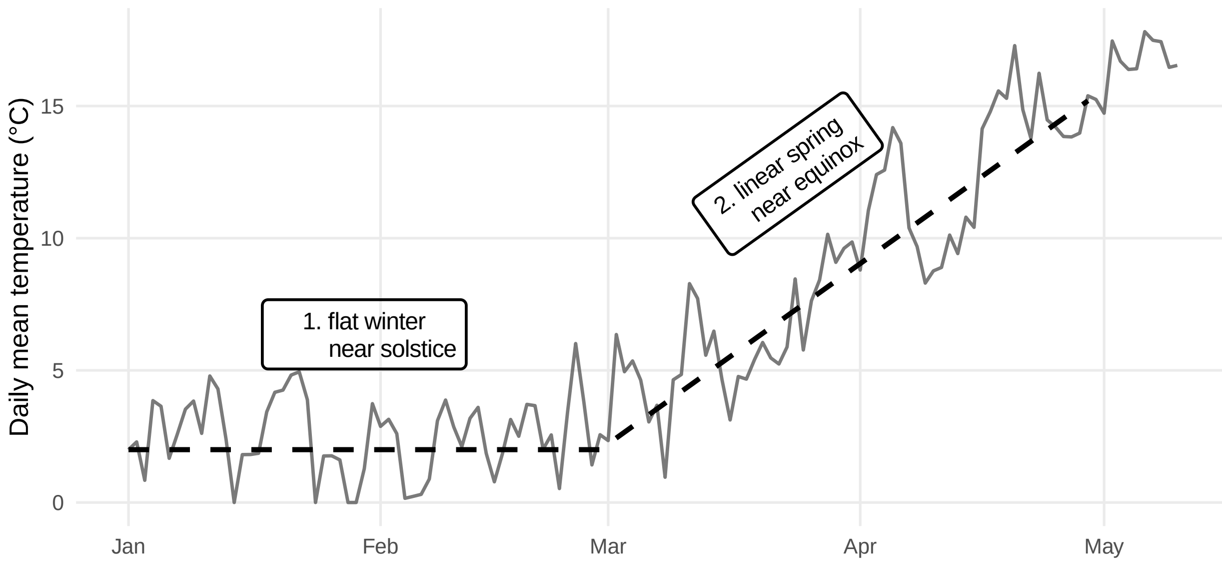

Figure 1: We model the daily mean temperature (solid line) as trend (dashed line) plus noise. The timing of springtime events in temperate climates depends on whether temperature is accumulated during one of two regimes, 1. the stationary regime (relatively flat near the winter solstice) or 2. the non-stationary regime (relatively linear near the spring equinox).

the influence of noise is strikingly different as reflected by whether t is in the numerator or the denominator. In the non-stationary regime—common near the equinox—t is in the denominator, and thus the variance is small when t is large. The interpretation is that noise is quickly overwhelmed by the rapid quadratic increase of within-season temperatures. This effectively acts to synchronize plants (and buds within plants) with the same threshold—making small differences in microclimate or other idiosyncratic effects negligible. In contrast, under the stationary regime—common near the solstice—t is in the numerator, and thus the variance is large when t is large. The interpretation is that noise can remain relatively influential so that small microclimate differences translate into larger differences in threshold-crossing times. This prediction fits well with experiments conducted under constant temperature regimes, which find increasing variance with decreasing baseline temperatures (Charrier:2011aa). In such experiments, buds often have extremely variable budburst timings, exactly as our model predicts (see Appendix).

The fact that the variance is predicted to be small in the non-stationary case and larger in the stationary case suggests the observed variance will increase when anthropogenic warming shifts accumulation from the non-stationary regime to the stationary regime. Indeed, as warming advances events earlier in the calendar, thresholds are reached in portions of the season with flatter mean temperatures where timing can be more sensitive to local variation. In such cases, warming produces a regime shift that simultaneously (i) advances springtime events potentially at a faster rate and (ii) reduces synchrony across nearby locations, producing a more variable and less coordinated onset of spring even as it arrives sooner, a phenomenon we refer to as a 'scattered spring.' ~~This regime shift produces a 'scattered spring' when climate change moves events earlier on average while simultaneously reducing coordination across individuals and locations, making spring less~~

~~tightly synchronized even as it arrives sooner.~~

Whether warming leads to a more synchronized or more scattered spring depends on the local joint evolution of (α, β) and on the threshold t relevant for a given species and life-cycle event. In some locations, climate change may manifest primarily as higher winter temperatures (an increase in α) with relatively little change in the rate at which temperatures rise during spring (a modest change in β). In others, winter temperatures may change less while spring transitions steepen (a larger increase in β). The stopped-random-walk framework offers a simple way to translate spatially varying climate change into spatially varying predictions about both the advancement of spring and the stability of ecosystem coordination. To demonstrate this, we examine data from the historical lilac–honeysuckle phenology network compiled by Rosemartin et al. (2015), which contains leafing and flowering dates collected across the continental United States from 1956–2014 for purple common lilac (*Syringa vulgaris*), a cloned lilac cultivar (*S.*,×,*chinensis* 'Red Rothomagensis'), and two cloned honeysuckle cultivars (*Lonicera tatarica* Arnold Red and *L. korolkowii* Zabeli).

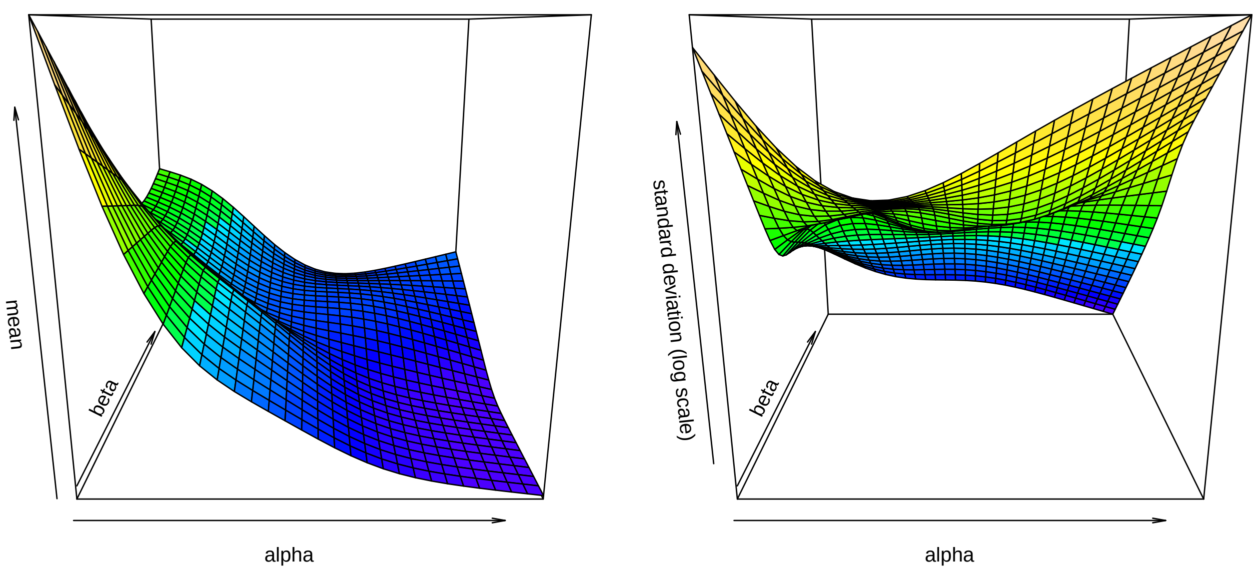

Figure 2: Mean and log standard deviation of lilac-honeysuckle bloom dates (n = 23,864 from Rosemartin et al. 2015) by baseline temperatures (α) and within-season temperature increases (β) estimated by a generalized additive model. The left figure shows the mean generally decreases at an increasing rate as α and β increase. The standard deviation can increase or decrease, again depending on the relative values of α and β.

We test whether increasing baseline winter and/or spring temperatures shift flowering times as predicted by our model in two ways. We first bin sites and years within a grid of α (rows) and β (columns) and calculate the mean and standard deviation of the observed bloom date (number of days since January 1) within each bin. We then estimate of the conditional mean and standard deviations nonparametrically using a generalized additive model (GAM), see Hastie and Tibshirani (1986) and Wood (2017). The estimated conditional means and standard deviations are displayed in Tables 1 and 2 (binning) and Figure 2 (GAM).

Both approaches (binning and GAM) show shifts in bloom dates across varying baseline winter and increasing spring temperatures generally supported our model predictions. First, holding α fixed, larger β is associated with earlier mean bloom dates, which generally advance at a decreasing rate. The same is true for holding β fixed, and increasing α. Holding α at a large value, the influence of β is considerably smaller. This is predicted by the model since, under such conditions, plants accumulate sufficient temperature under the stationary regime in which β plays no role, and thus bloom before reaching the the non-stationary regime in which β plays a role.

Consistent with the theory, the standard deviations (and thus the variances) generally declined with higher increasing spring temperatures (i.e., as β increases holding α fixed). (In the right panel of Figure 2, note that the yellow region corresponds with small β and the blue region corresponds with large β.) This occurs because when baseline temperatures (α) are held fixed, the percent of buds that bloom in the stationary climate is unchanged, and the buds that makes it to the increasing temperature regime have a lower standard deviation. However, holding β fixed and increasing α produces a regime shift in which more buds bloom under the stationary climate regime and thus the bloom date dispersion increases. This is the "scattered spring" predicted from our model. Warmer winters advance bloom dates in to an earlier portion of the seasonal cycle in which forcing is effectively flatter and cumulative temperatures grow more slowly.

We conclude that the basic cumulative sum model explains the observed phenomenon in which the advancement of phenological events are slowing as climates warm and the variance is in some cases increasing and in others decreasing. In particular, the model predicts a general desynchronization of springtime events as their timing moves into the stationary regime closer to the solstice. Though our model is simple, it does not exclude the possibility of a more complex phenomenon, such as chilling. Indeed, a similar phenomenon would occur if we allowed parameters to vary according to winter temperatures or other factors such as photoperiod. In addition, we could also allow additional parameters such as the day in which accumulation starts or specify a more general function of temperature that plants accumulate (which may vary within a day). We could also allow the idiosyncratic variation to be correlated across time or across neighboring processes. However, these complications are not necessary to explain slowing responses or increased variance. The stopping-time framework shows these observations follow from the cumulative sum model that has been used to predict bloom dates for over two centuries.

Methods

Our data analysis considers all sites in the historical lilac–honeysuckle phenology network that recorded the phenophase full bloom and were within 10 miles of a GHCND site with complete temperature records, resulting in $23,864$ observations. For each observation, we estimate α using the average daily temperature between January and February. We estimate β using the slope of an linear regression model fit by regressing daily temperature on day index over March and April. Temperatures below 0 are set to 0. The parameters α and β are shown for nine random sites in Figures 4 and 5 in Appendix A.2, which have the same interpretation

Table 1: Mean bloom date (day-of-year) from lilac-honeysuckle data (Rosemartin et al. 2015) by average Jan-Feb temperature α (rows) and average Mar-Apr temperature increase β (columns).

$\alpha \setminus \beta$	$[-0.28, 0.07]$	$(0.07, 0.12]$	$(0.12, 0.16]$	$(0.16, 0.68]$
(0, 0.6]	163.05	154.24	148.96	142.56
(0.6, 1.6]	153.21	147.97	142.60	136.78
(1.6, 4]	138.86	133.56	130.06	126.34
(4, 16.4]	105.13	106.46	105.26	102.67

Table 2: Standard deviation of bloom date (day-of-year) from lilac-honeysuckle data (Rosemartin et al. 2015) by winter level α (rows) and spring warming rate β (columns).

$\alpha \setminus \beta$	$[-0.28, 0.07]$	$(0.07, 0.12]$	$(0.12, 0.16]$	$(0.16, 0.68]$
(0, 0.6]	16.08	12.36	10.44	9.61
(0.6, 1.6]	15.24	12.87	13.20	12.67
(1.6, 4]	17.97	16.52	16.31	14.56
(4, 16.4]	22.78	19.84	17.23	14.76

as Figure 1. Site 14726 in New Mexico (Year 1969) is a good example of a bloom in regime 2 since relatively little temperature was accumulated during January and February. The average bloom date was May 8th with a standard deviation of 17 days. Site 14276 in New York (Year 1982) is a good example of a bloom between regimes 1 and 2, which had an average bloom date of March 18 with a standard deviation of 27.

The mean and standard deviation are shown for a grid of α and β by quartiles in Tables 1 and 2. We also fit a two-parameter Gaussian location–scale model using the `gaulss` function from `mgcv` where

$$\mu(\alpha, \beta) = \mathbb{E}[\nu(\tau) \mid \alpha, \beta] \quad \text{and} \quad \sigma(\alpha, \beta) = \mathrm{SD}(\nu(\tau) \mid \alpha, \beta).$$

Both μ and σ are modeled with tensor-product splines with basis dimension k = 10. The fitted functions are shown in Figure 2.

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Summary of analytical solution

To fix ideas, consider the day a bud blooms at a given location. We model the effective temperature experienced by that bud on day i by a linear trend plus noise,

$$X_i = \alpha + \beta\, i + \epsilon_i, \tag{1}$$

where $\alpha > 0$ is a baseline level, $\beta \geq 0$ is a within-season warming rate, and $\{\epsilon_i\}_{i\geq 1}$ are independent and identically distributed with $\mathbb{E}[\epsilon_i] = 0$ and $\mathrm{Var}(\epsilon_i) = \sigma^2$, representing idiosyncratic variation.

Let

$$Z_n \;=\; \sum_{i=1}^{n} X_i \qquad \text{and} \qquad \nu(t) \;=\; \min\{m : Z_m > t\} \tag{2}$$

denote cumulative temperature by day n and the observed bloom date, respectively, where $t > 0$ is a plant-specific tempearture-sum threshold. Thus $\nu(t)$ is the hitting time at which cumulative temperatures first exceed the threshold.

Standard results for stopped random walks yield asymptotic normal approximations for $\nu(t)$, with behavior governed by the parameters α and β. We consider two cases.

Regime 1: constant drift ($\beta = 0$) When average temperatures are positive and approximately constant,

$$\nu(t) \mathrel{\dot\sim} \mathrm{Normal}\!\left(\frac{t}{\alpha},\ \frac{\sigma^2\, t}{\alpha^3}\right), \tag{3}$$

see Gut (2011) and Gut (2009, chap. 3).

Regime 2: increasing drift ($\beta > 0$) When average temperatures increase, the hitting time satisfies

$$\nu(t) \mathrel{\dot\sim} \mathrm{Normal}\!\left(\sqrt{\frac{2t}{\beta}} - \frac{\alpha}{\beta} + \frac{1}{2},\ \frac{\sigma^2}{\beta^{3/2}\sqrt{2t}}\right), \tag{4}$$

with the derivation sketched in Appendix A.1 and simulations confirming the derivations in Appendix A.3. Note that when t is sufficiently large, the mean of $\nu(t)$ is approximately $\sqrt{2t\,/\,\beta}$.

The functional central limit theorem and linearization argument underlying these results hold somewhat generally, and thus similar limits can be derived when the $\{\epsilon_i\}_{i\geq 1}$ are weakly correlated or the temperature trajectory is not linear but sufficiently smooth.

From equations (3)–(4), we make two observations. Finding 1: there is a different nonlinear relationship between warming and the mean timing of springtime events, and thus warming can advance springtime events at an increasing rate. Finding 2: the two regimes are not equally sensitive to the idiosyncratic variation, and thus warming can increase the variation of springtime events, thereby scattering spring. We now discuss both findings in greater detail.

2.1 Finding 1: warming can advance springtime events at an increasing rate

In both regimes, the expected bloom date is a nonlinear function of the warming parameters.

Regime 1: constant drift ($\beta = 0$) From (3),

$$\mathbb{E}[\nu(t)] \approx \frac{t}{\alpha}, \qquad \mathrm{Var}(\nu(t)) \approx \frac{\sigma^2\, t}{\alpha^3}. \tag{5}$$

Increasing baseline temperatures (larger α) advances the expected bloom date, but the marginal advance diminishes since

$$\frac{\partial}{\partial \alpha}\, \mathbb{E}[\nu(t)] \approx -\frac{t}{\alpha^2}, \tag{6}$$

where the magnitude decreases as α grows.

Regime 2: increasing drift ($\beta > 0$) From (4), when t is large,

$$\mathbb{E}[\nu(t)] \approx \sqrt{\frac{2t}{\beta}}, \qquad \mathrm{Var}(\nu(t)) \approx \frac{\sigma^2}{\beta^{3/2}\sqrt{2t}}. \tag{7}$$

Increasing the within-regime warming rate (larger β) also advances the expected bloom date with diminishing returns since

$$\frac{\partial}{\partial \beta}\, \mathbb{E}[\nu(t)] \approx -\frac{1}{2}\sqrt{\frac{2t}{\beta^3}}. \tag{8}$$

However, if warming climates shift the bloom date from regime 2 to regime 1, then the rate of advancement can increase, particularly if increases in α outpace increases in β.

2.2 Finding 2: warming can scatter springtime events

Equations (3)–(4) also clarify how warming affects dispersion. Holding the regime fixed, warming reduces variance. That is, $\mathrm{Var}(\nu(t))$ decreases with α in (3) and with β in (4). However, just as in Section 2.1, warming can change which regime is relevant at the realized bloom date. When an event is advanced sufficiently early, idiosyncratic variation plays a larger role in determining the hitting time.

One way to quantify sensitivity to idiosyncratic variation is through the (squared) coefficient of variation,

$$\mathrm{CV}^2 \;=\; \frac{\mathrm{Var}(\nu(t))}{\mathbb{E}[\nu(t)]^2}.$$

In regime 1, $\mathbb{E}[\nu(t)] \approx t/\alpha$ and $\mathrm{Var}(\nu(t)) \approx \sigma^2 t/\alpha^3$, so

$$\mathrm{CV}_1^2 \;=\; \frac{\mathrm{Var}(\nu(t))}{\mathbb{E}[\nu(t)]^2} \;\approx\; \frac{\sigma^2 t/\alpha^3}{(t/\alpha)^2} \;=\; \frac{\sigma^2}{\alpha\, t}.$$

In regime 2, $\mathbb{E}[\nu(t)] \approx \sqrt{2t/\beta}$ and $\mathrm{Var}(\nu(t)) \approx \sigma^2/(\beta^{3/2}\sqrt{2t})$, hence

$$\mathrm{CV}_2^2 \;=\; \frac{\mathrm{Var}(\nu(t))}{\mathbb{E}[\nu(t)]^2} \;\approx\; \frac{\sigma^2/(\beta^{3/2}\sqrt{2t})}{(2t/\beta)} \;=\; \frac{\sigma^2}{\sqrt{\beta}\,(2t)^{3/2}}.$$

Both expressions increase when deterministic accumulation is weaker relative to σ (e.g., smaller α and/or smaller effective β). Consequently, warming tends to reduce variability within a given regime by strengthening deterministic forcing.

To compare regimes, note that

$$\frac{\mathrm{CV}_1^2}{\mathrm{CV}_2^2} \;\approx\; \frac{\sigma^2/(\alpha t)}{\sigma^2/(\sqrt{\beta}\,(2t)^{3/2})} \;=\; \frac{\sqrt{\beta}\,2^{3/2}}{\alpha}\,\sqrt{t}.$$

Thus, since α and β are nearly always greater than 0, the regime 1 becomes arbitrarily more sensitive to microclimate fluctuations than regime 1 for large t. This produces a more variable, less synchronized and thus "scattered" spring when warming shifts threshold crossings from regime 2 to regime 1.

Experimental Data

We first consider a controlled experiment on walnut trees (*Juglans*~sp.) conducted by Charrier et al. (2011). The data examines 30 trees comprising 6 genotypes at 2 locations. In November of the study year, stems were sampled from each tree and cut into 7 centimeter segments containing a single bud. All segments were chilled at 4°C and then transferred to constant 'forcing' environments with temperatures set at one of $5, 10, 15, 20$, or 25°C. For each temperature, the outcome is the response time, the time (in days) until budburst determined from the date at which 50% of buds reached stage 15 of the BBCH scale.

This experiment isolates the winter regime because temperature is held constant over time. Recall that in such cases, cumulative forcing grows approximately linearly,

$$Z_n \approx \alpha n + \sum_{i=1}^{n} \epsilon_i,$$

so that the hitting time $\nu(t)$ satisfies the approximation

$$\nu(t) \mathrel{\dot\sim} \mathrm{Normal}\left(\frac{t}{\alpha},\ \frac{\sigma^2 t}{\alpha^3}\right),$$

implying that increasing α should (i) reduce the mean time to budburst and (ii) reduce dispersion. In addition, the squared coefficient of variation,

$$\mathrm{CV}^2 = \frac{\mathrm{Var}(\nu(t))}{\mathbb{E}[\nu(t)]^2},$$

provides a unit-free measure of sensitivity to stochastic fluctuations, which should decrease as α increases.

Table 3: Summary data from the Walnut forcing experiment (Charrier et al. 2011).

α	n	mean	sd	cv^2
5	30	178	27.28	0.0236
10	30	88	12.44	0.0199
15	30	39	11.33	0.0831
20	30	26	7.67	0.0851
25	30	20	4.45	0.0480

Table 3 reports the mean and standard deviation of the response times across the 30 stems at each forcing temperature, along with the sensitivity measure CV^2. The pattern is consistent with the regime 1 model. Higher forcing temperatures advance budburst at a decreasing rate (the mean falls with α), and the dispersion of budburst timing is smaller at warmer forcing temperatures. Thus, in an experimental setting, the stopped-random-walk approximation captures the basic comparative statics of constant-temperature forcing on both the level and variability of budburst times. (The sensitivty does decline from $\alpha = 20$ to $\alpha = 25$. The reason for low sensitivity for $\alpha = 5$ and 10 is unclear and may reflect censoring.)

A. Appendix

A.1 Sketch of Theoretical Results

We follow the notation in Gut (2009, chap. 6). Let $X_i = \alpha + \beta i + \epsilon_i$ denote the temperature on day i, where $\alpha > 0$, $\beta \geq 0$, and the $\{\epsilon_i\}$ are independent and identically distributed with $\mathbb{E}[\epsilon_i] = 0$, and $\mathrm{Var}(\epsilon_i) = \sigma^2$. Define the deterministic cumulative sum

$$\xi_n = \sum_{i=1}^{n} \mathbb{E}[X_i] = \alpha n + \frac{\beta}{2} n(n+1) = \frac{\beta}{2} n^2 + \beta\gamma n \quad \text{where} \quad \gamma = \frac{\alpha}{\beta} + \frac{1}{2}$$

and $\xi_n = \alpha n$ when $\beta = 0$. Denote the sum of the stochastic component $S_n = \sum_{i=1}^{n} \epsilon_i$. Our object of interest is the first-passage time of $Z_n = S_n + \xi_n$,

$$\nu(t) = \min\{n \geq 1 : \ Z_n > t\}.$$

In the language of Lai and Siegmund (1977), $\{Z_n\}$ is a perturbed random walk: the random walk $\{S_n\}$ is perturbed by a deterministic term $\{\xi_n\}$. This setup yields two asymptotic regimes, corresponding to the empirical distinction between regime 1 ($\beta \approx 0$) and regime 2 ($\beta > 0$), which in turn underlies our qualitative predictions about nonlinear changes in themean and variance.

In regime 1, $\beta = 0$, and we apply the usual asymptotic argument for the hiting time of a random walk with constant positive drift as $t \to \infty$,

$$\nu(t) \mathrel{\dot\sim} \mathrm{Normal}\left(\frac{t}{\alpha}, \ \frac{\sigma^2 t}{\alpha^3}\right),$$

see Gut (2011) and Gut (2009, chap. 3).

In regime 2, $\beta > 0$, and the deterministic component $\xi_n = \frac{\beta}{2} n^2 + \beta\gamma n$ grows quadratically so that the threshold t is reached near the deterministic crossing time $m(t)$ in which $\xi_{m(t)} \approx t$, i.e.

$$m(t) = \frac{-\beta\gamma + \sqrt{\beta^2\gamma^2 + 2\beta t}}{\beta} = \sqrt{\frac{2t}{\beta}} - \gamma + o(1).$$

By the functional central limit theorem, the deviation of $\nu(t)$ about $m(t)$ is approximately normal. Linearization of $Z_{\nu(t)} \approx t$ around $m(t)$ yields

$$\nu(t) \mathrel{\dot\sim} \mathrm{Normal}\left(m(t), \ \frac{\sigma^2\, m(t)}{\big(\alpha + \beta\, m(t)\big)^2}\right) \ = \ \mathrm{Normal}\left(\sqrt{\frac{2t}{\beta}} - \gamma, \ \frac{\sigma^2}{\beta^{3/2}\sqrt{2t}}\right),$$

where we have used the fact that $\alpha + \beta m(t) \sim \beta m(t) \sim \sqrt{2\beta t}$ for $t \gg 0$. Thus, in regime 2 the mean hitting time scales like $\sqrt{t}$ (rather than linearly in t) and the variance shrinks at rate $t^{-1/2}$ (rather than growing at t).

A.2 Additional Figures

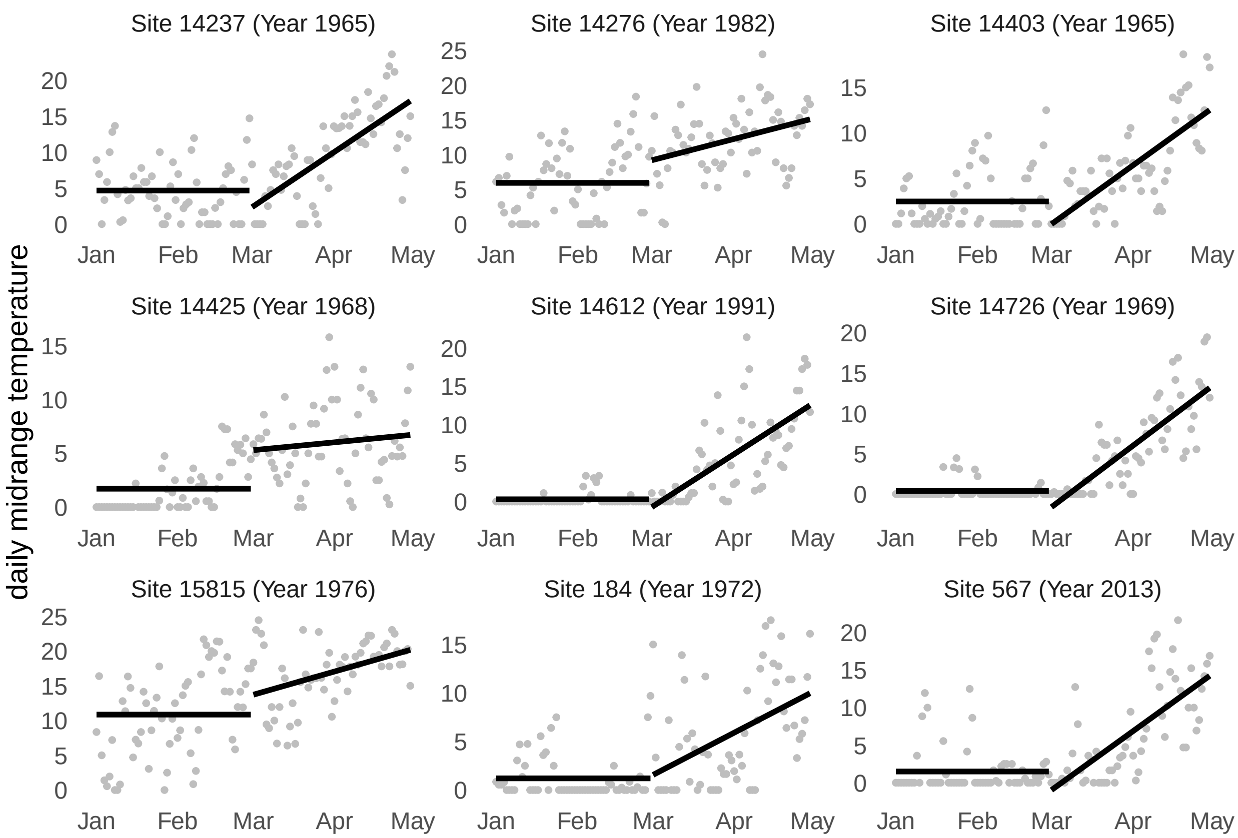

Figure 3: Springtime daily midrange temperatures (points) with α (mean from Jan to Feb) and β (slope from Mar to Apr) for nine randomly selected sites from lilac-honeysuckle dataset (Rosemartin et al. 2015) as measured by GHCND stations nearby. Temperatures below 0 are set to 0.

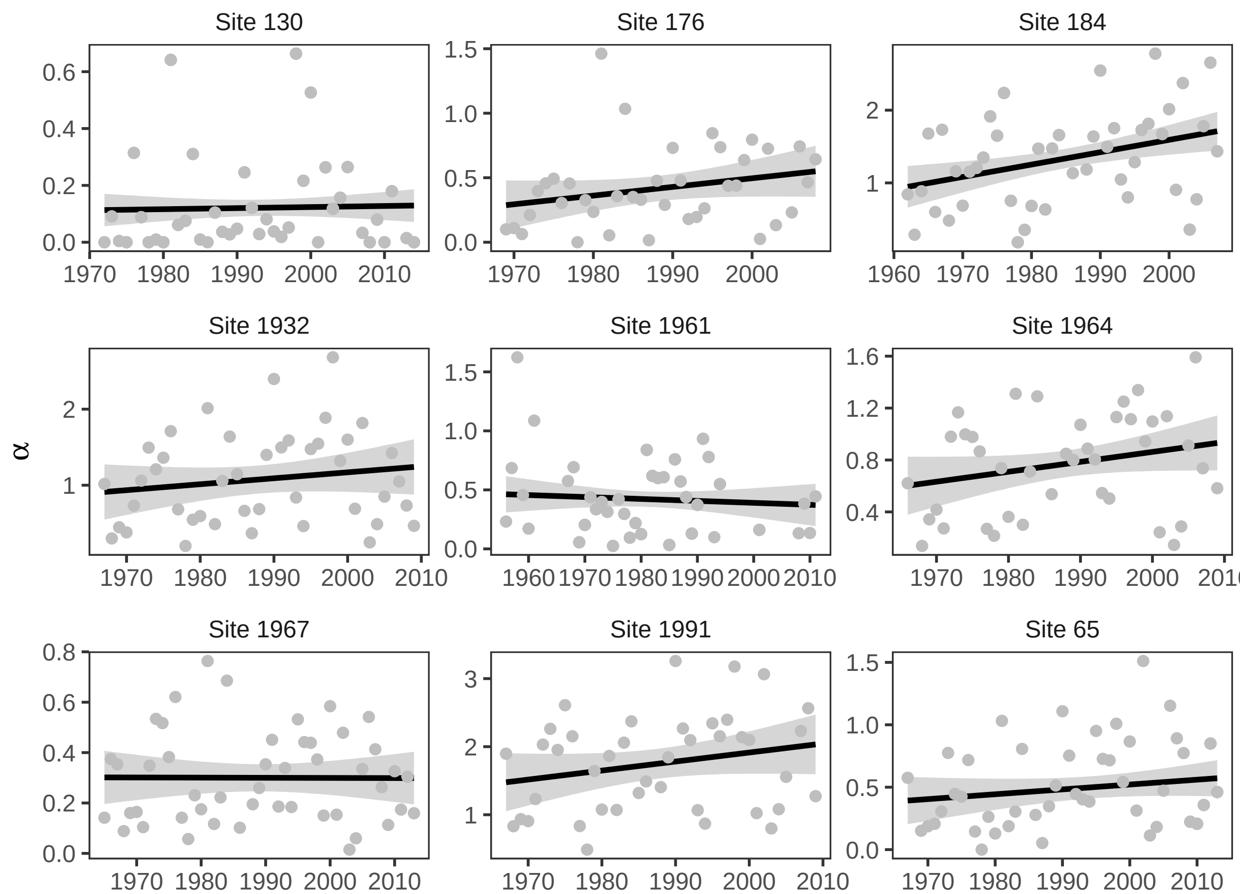

Figure 4: Changes in α for the nine sites from lilac-honeysuckle data (Rosemartin et al. 2015) with longest running records as as measured by GHCND stations nearby.

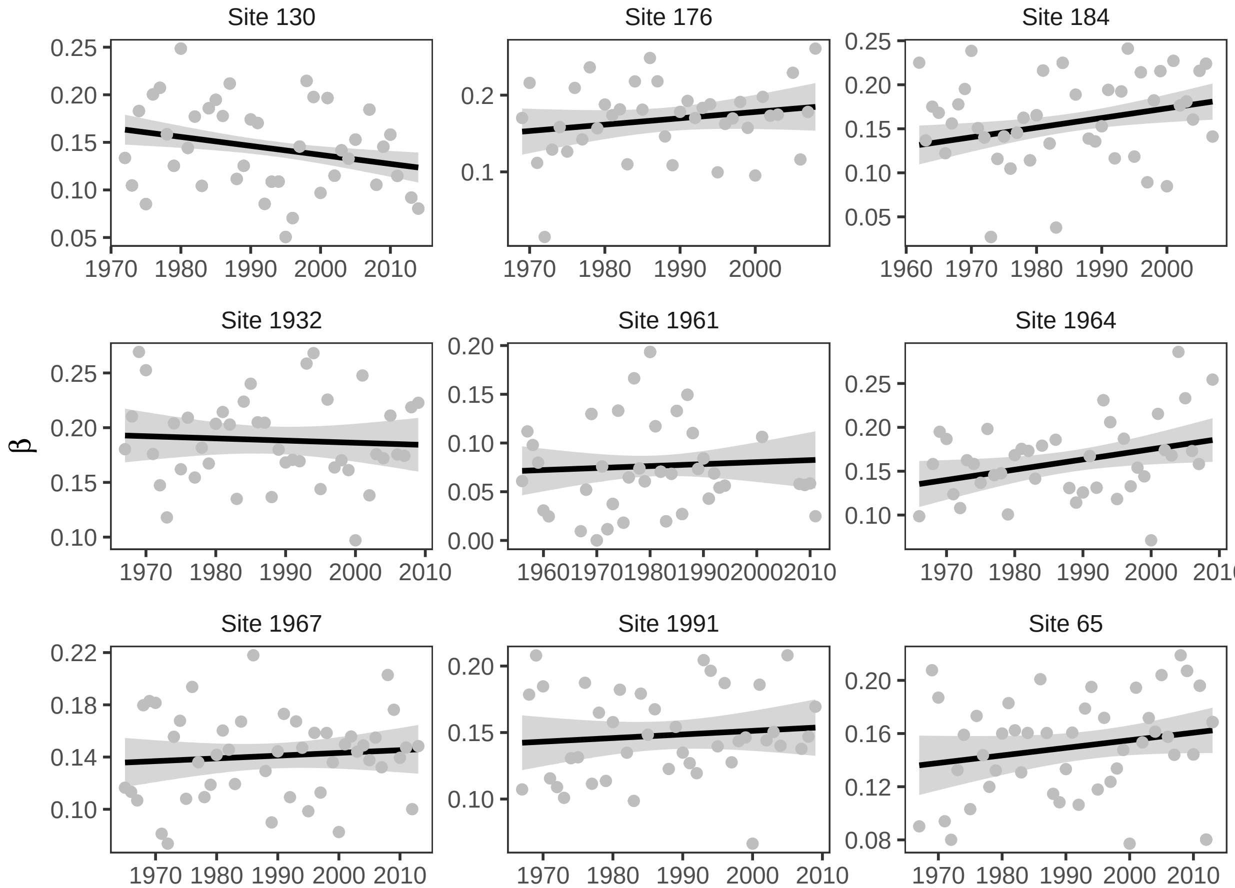

Figure 5: Changes in β for the nine sites from lilac-honeysuckle data (Rosemartin et al. 2015) with longest running records as as measured by GHCND stations nearby.

A.3 Simulations

We run two simulations to study the results from Sections 2 and 3. The first checks the asymptotic approximation from Section 2. The second checks the interpretation of the data from Section 3.

Simulation 1

The first simulation verifies the accuracy of the two asymptotic normal approximations for the hitting time

$$\nu(t) = \min\left\{n \geq 1 : \sum_{i=1}^{n} X_i > t\right\}$$

under the model

$$X_i = \alpha + \beta i + \varepsilon_i, \qquad \varepsilon_i \sim \text{Normal}(0, \sigma^2).$$

For simulation, we set $\sigma = 20$ and generate daily temperatures until the threshold t is reached, recording the corresponding hitting time $\nu(t)$. We repeat this procedure $R = 10{,}000$ times for thresholds $t = \{1000, 2000\}$, $\alpha = \{2, 4\}$ and $\beta = \{0, 0.1\}$. When $\beta = 0$ the simulation matches the winter regimes, and when $\beta = 0.1$, it matches the spring regime. The simulated distribution of $\nu(t)$ is represented in Figure 6 below by the histograms.

We then compare these simulations with the theoretical approximations, we standardized each simulated hitting time using the relevant asymptotic mean and standard deviation. When $\beta = 0$, we use the winter regime approximation

$$\nu(t) \mathrel{\dot\sim} \text{Normal}\left(\frac{t}{\alpha},\ \frac{\sigma^2 t}{\alpha^3}\right).$$

When $\beta > 0$, we use the spring regime approximation

$$\nu(t) \mathrel{\dot\sim} \text{Normal}\left(\sqrt{\frac{2t}{\beta}} - \frac{\alpha}{\beta} + \frac{1}{2},\ \frac{\sigma^2}{\beta^{3/2}\sqrt{2t}}\right).$$

For each setting we computed the standardized variable

$$z = \frac{\nu(t) - \mu}{\text{sd}},$$

plotted its histogram, and overlaid the standard normal density. Across both regimes, the standardized histograms align closely with the Normal$(0, 1)$ curve, and the agreement improves as t increases, providing a visual confirmation of the derived asymptotic distributions.

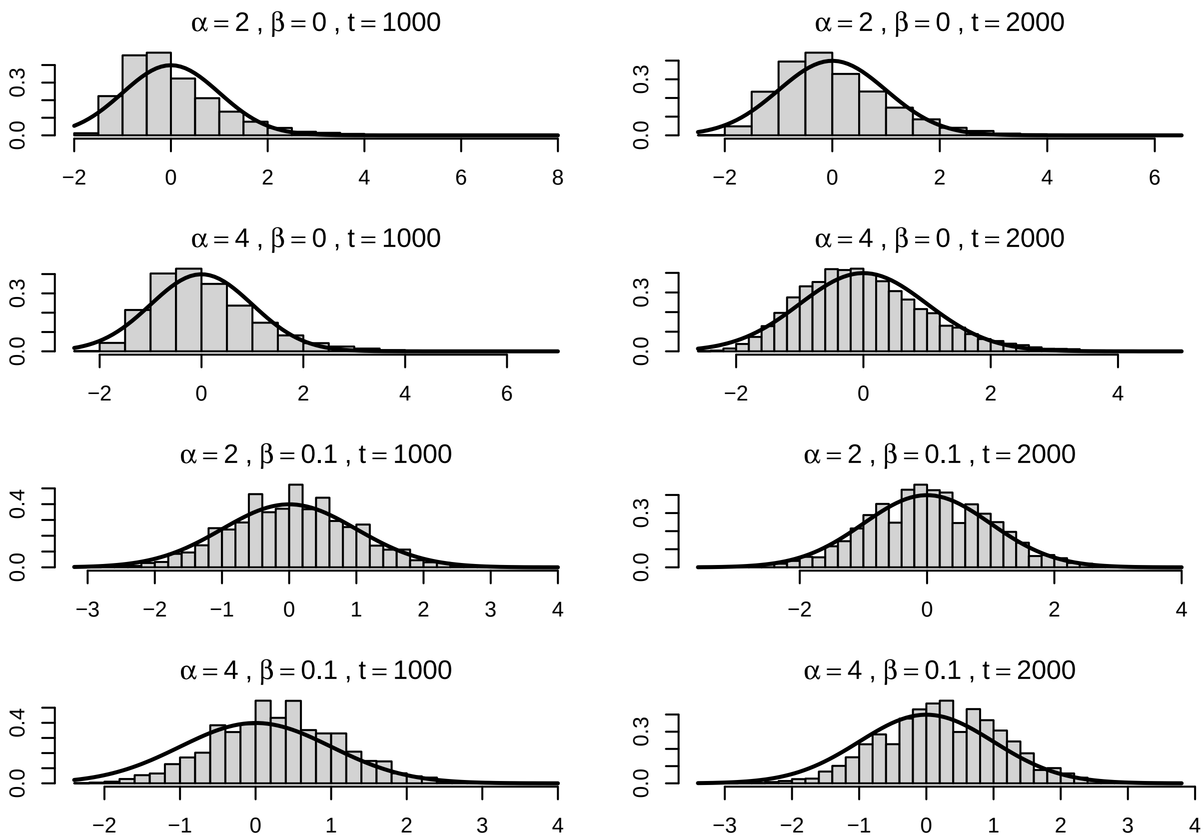

Figure 6: Simulations comparing the distribution of bloom dates under two regimes: stationary (top) and non-stationary (bottom).

Simulation 2

To verify our interpretation of the data in Section 3, we simulate the model from section. In each replicate we generated daily temperatures

$$X_i = \mu_i + \varepsilon_i, \qquad \varepsilon_i \sim \mathrm{Normal}(0, \sigma^2),$$

with $\sigma = 20$, where the deterministic component follows

$$\mu_i = \begin{cases} \alpha, & i = 1, \ldots, 90 \quad \text{(January–February)}, \\ \alpha + \beta(i - 90), & i = 91, \ldots, 180 \quad \text{(March–April)}. \end{cases}$$

Bloom occurs when cumulative temperature first exceeds a threshold t, i.e.

$$\nu(t) = \min\Big\{ n \geq 1 : \sum_{i=1}^{n} X_i > t \Big\}.$$

We simulated $R = 10{,}000$ independent realizations of $\nu(t)$ for each combination of $\alpha \in \{4, 8, 10\}$ and $\beta \in \{0.2, 0.4, 0.8\}$, and for thresholds $t \in \{1000, 2000\}$. For each parameter pair we report the Monte Carlo mean and standard deviation of $\nu(t)$ (Tables 4a–4d), arranged with rows indexing α and columns indexing β.

The results reproduce the qualitative patterns observed in the lilac application. Mean bloom dates decrease as either α increases or β increases (Tables~4a and 4c). Within a fixed α row, increasing β reduces the standard deviation of bloom dates (Tables~4b and 4d), consistent with the prediction that that increasing average temperatures diminish the influence of noise on the threshold-crossing time. Holding β fixed, the standard deviation is often larger at higher α (particularly for $\beta \in \{0.4, 0.8\}$), consistent with the regime-shift intuition that earlier crossings can occur in a flatter portion of the seasonal cycle where microclimate variability has greater influence.

Table 4: Simulated mean and standard deviation of the bloom date $\nu(t)$ across $R = 10{,}000$ Monte Carlo replicates, for thresholds $t \in \{1000, 2000\}$. Rows index the winter level α and columns index the spring warming rate β.

(a) mean, $t = 1000$

$\alpha \setminus \beta$	0.2	0.4	0.8
4	151.66	136.83	124.86
8	114.87	111.27	106.85
10	98.45	97.20	95.72

(b) sd, $t = 1000$

$\alpha \setminus \beta$	0.2	0.4	0.8
4	15.48	10.42	7.21
8	16.53	13.27	10.50
10	15.94	14.45	12.71

(c) mean, $t = 2000$

$\alpha \setminus \beta$	0.2	0.4	0.8
4	199.54	171.15	149.16
8	169.81	152.43	137.37
10	156.22	143.40	131.34

(d) sd, $t = 2000$

$\alpha \setminus \beta$	0.2	0.4	0.8
4	10.96	7.22	4.84
8	10.93	7.50	5.11
10	10.69	7.66	5.36